

1. CO3-AT2 - Case Study.

Student Record Retrieval System

A university database stores student details in sorted order. Analyze how binary search improves retrieval efficiency. Identify requirements, apply divide and conquer, analyze complexity and justify its effectiveness.

Binary Search:

{ 101, 105, 110, 115, 120, 125, 130 }.

$$K = 115$$

$$\text{mid} = \frac{l+h}{2} \Rightarrow \frac{0+6}{2} = 3.$$

$$\text{mid} = 3.$$

if $a[\text{mid}] == k$:

$$115 = 115$$

print (found)

Time complexity = $\log(n)$

Space complexity = $O(\log n)$